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Studierichting: Informatica
Oefeningenreeks 1E nr. 14

We delen $z^4 + z^2 + 1$

door $(z - i)(z + 2i)(z - 1)$

en noemen de rest $R(z)$

Omdat de deler graad 3 heeft, heeft de rest graad hoogstens 2

We stellen dus $R(z) = az^2 + bz + c$

$$R(i) = i^4 + i^2 + 1 = 1$$

$$R(-2i) = (-2i)^4 + (-2i)^2 + 1 = 13$$

$$R(1) = 1^4 + 1^2 + 1 = 3$$

$$\begin{cases} -ai + bi + c = 1 \\ -4a - 2bi + c = 13 \\ a + b + c = 3 \end{cases}$$
$$\Rightarrow \begin{cases} -a + bi + c = 1 \\ -4a - 2bi + c = 13 \\ a + b + c = 3 \end{cases}$$

$$\text{vgl2} = 2\text{vgl1} + \text{vgl2}$$

$$\Rightarrow \begin{cases} -6a + 3c = 15 \\ -4a - 2bi + c = 13 \\ a + b + c = 3 \end{cases}$$

$$\text{vgl1} = \frac{\text{vgl1}}{3}$$

$$\Rightarrow \begin{cases} -2a + c = 5 \\ -4a - 2bi + c = 13 \\ a + b + c = 3 \end{cases}$$

$$\text{vgl2} = \text{vgl2} - \text{vgl1}$$

$$\Rightarrow \begin{cases} -2a + c = 5 \\ -2a - 2bi = 8 \\ a + b + c = 3 \end{cases}$$

$$\text{vgl2} = \frac{\text{vgl2}}{-2}$$

$$\Rightarrow \begin{cases} -2a + c = 5 \\ a + bi = -4 \\ a + b + c = 3 \end{cases}$$

Uit vgl1 volgt $c = 5 + 2a$

Invullen in vgl3 geeft $a + b + (5 + 2a) = 3$

$$3a + b = -2$$

$$b = -2 - 3a$$

Uit vgl2 volgt $bi = -4 - a$

$$b = (-4 - a) \frac{1}{i}$$

$$b = (-4 - a)(-i)$$

$$b = (4 + a)i$$

Dus $-2 - 3a = (4 + a)i$

$$-2 + a(-3 - i) = 4i$$

$$a(-3 - i) = 2 + 4i$$

$$a = \frac{2 + 4i}{-3 - i}$$

$$a = \frac{(2 + 4i)(-3 + i)}{(-3 - i)(-3 + i)}$$

$$a = \frac{-10 - 10i}{10}$$

$$a = -1 - i$$

$$b = -2 - 3a = -2 - 3(-1 - i) = 1 + 3i$$

$$c = 5 + 2a = 5 + 2(-1 - i) = 3 - 2i$$

$$\Rightarrow R(z) = (-1 - i)z^2 + (1 + 3i)z + (3 - 2i)$$

References